

```
!pip install transformers datasets torch tqdm matplotlib
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: transformers in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (4.49.0)

Requirement already satisfied: datasets in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (3.3.2)

Requirement already satisfied: torch in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (2.6.0)

Requirement already satisfied: tqdm in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (4.67.1)

Requirement already satisfied: matplotlib in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (3.10.1)

Requirement already satisfied: filelock in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (3.17.0)

Requirement already satisfied: huggingface-hub<1.0,>=0.26.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.29.2)

Requirement already satisfied: numpy>=1.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (2.2.3)

Requirement already satisfied: packaging>=20.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (24.2)

Requirement already satisfied: pyyaml>=5.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (6.0.2)

Requirement already satisfied: regex!=2019.12.17 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (2024.11.6)

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Requirement already satisfied: tokenizers<0.22,>=0.21 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.21.0)

Requirement already satisfied: safetensors>=0.4.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from transformers) (0.5.3)

Requirement already satisfied: pyarrow>=15.0.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (19.0.1)

Requirement already satisfied: dill<0.3.9,>=0.3.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from datasets) (0.3.8)

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Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)

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Requirement already satisfied: setuptools in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from torch) (75.8.2)

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Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from sympy==1.13.1->torch) (1.3.0)

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Requirement already satisfied: fonttools>=4.22.0 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (4.56.0)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (1.4.8)

Requirement already satisfied: pillow>=8 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from matplotlib) (11.1.0)

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Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

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Requirement already satisfied: tzdata>=2022.7 in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from pandas->datasets) (2025.1)

[notice] A new release of pip is available: 24.3.1 -> 25.0.1

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```
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```
pip install datasets
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Requirement already satisfied: filelock in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.17.0)

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Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from pandas->datasets) (2.9.0.post0)
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Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
import torch
import numpy as np
from transformers import GPT2Tokenizer, GPT2LMHeadModel, Trainer, TrainingArguments
from datasets import load_dataset
from torch.utils.data import DataLoader
from transformers import DataCollatorForLanguageModeling
```

```

# Check GPU availability
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

c:\Users\CSIR_prosthetic\AppData\Local\Programs\Python\Python313\Lib\
site-packages\tqdm\auto.py:21: TqdmWarning: IPProgress not found.
Please update jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
    from .autonotebook import tqdm as notebook_tqdm

Using device: cuda

dataset = load_dataset("wikitext", "wikitext-2-raw-v1")

# Split dataset into train and test
train_texts = dataset["train"]["text"]
test_texts = dataset["test"]["text"]

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
tokenizer.pad_token = tokenizer.eos_token # Set pad token

model = GPT2LMHeadModel.from_pretrained("gpt2").to(device)

def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length",
truncation=True, max_length=512)

train_encodings = dataset["train"].map(tokenize_function,
batched=True)
test_encodings = dataset["test"].map(tokenize_function, batched=True)

data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer,
mlm=False)

train_dataloader = DataLoader(train_encodings, batch_size=8,
shuffle=True, collate_fn=data_collator)
test_dataloader = DataLoader(test_encodings, batch_size=8,
shuffle=False, collate_fn=data_collator)

!pip install -U accelerate transformers[torch]

Defaulting to user installation because normal site-packages is not
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appdata\roaming\python\python312\site-packages (1.4.0)
Requirement already satisfied: transformers[torch] in c:\users\
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(4.49.0)
Requirement already satisfied: numpy<3.0.0,>=1.17 in c:\users\
csir_prosthetic\appdata\roaming\python\python312\site-packages (from
accelerate) (2.2.3)

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Requirement already satisfied: psutil in c:\users\csir_prosthetic\appdata\roaming\python\python312\site-packages (from accelerate) (7.0.0)

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Requirement already satisfied: torch in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (2.6.0+cu124)
Requirement already satisfied: accelerate in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (1.4.0)
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Requirement already satisfied: requests in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (2.32.3)

Requirement already satisfied: tokenizers<0.22,>=0.21 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (0.21.0)

Requirement already satisfied: safetensors>=0.4.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (0.5.3)

Requirement already satisfied: tqdm>=4.27 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from transformers) (4.67.1)

Requirement already satisfied: pyarrow>=15.0.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (19.0.1)

Requirement already satisfied: dill<0.3.9,>=0.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.3.8)

Requirement already satisfied: pandas in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (2.2.3)

Requirement already satisfied: xxhash in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.5.0)

Requirement already satisfied: multiprocessing<0.70.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (0.70.16)

Requirement already satisfied: fsspec<=2024.12.0,>=2023.1.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from fsspec[http]<=2024.12.0,>=2023.1.0->datasets) (2024.12.0)

Requirement already satisfied: aiohttp in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from datasets) (3.11.13)

Requirement already satisfied: typing-extensions>=4.10.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (4.12.2)

Requirement already satisfied: networkx in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (3.4.2)

Requirement already satisfied: jinja2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (3.1.6)

Requirement already satisfied: setuptools in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (75.8.2)

Requirement already satisfied: sympy==1.13.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from torch) (1.13.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from sympy==1.13.1->torch) (1.3.0)

Requirement already satisfied: psutil in c:\users\csir_prosthetic\appdata\roaming\python\python313\site-packages (from accelerate) (7.0.0)

Requirement already satisfied: aiohappyeyeballs>=2.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (2.5.0)

Requirement already satisfied: aiosignal>=1.1.2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.3.2)

Requirement already satisfied: attrs>=17.3.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (25.1.0)

Requirement already satisfied: frozenlist>=1.1.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.5.0)

Requirement already satisfied: multidict<7.0,>=4.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (6.1.0)

Requirement already satisfied: propcache>=0.2.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (0.3.0)

Requirement already satisfied: yarl<2.0,>=1.17.0 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from aiohttp->datasets) (1.18.3)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.4.1)

Requirement already satisfied: idna<4,>=2.5 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\csir_prosthetic\appdata\local\programs\python\python313\lib\site-packages (from requests->transformers) (2025.1.31)

Requirement already satisfied: colorama in c:\users\csir_prosthetic\

```
appdata\roaming\python\python313\site-packages (from tqdm>=4.27-
>transformers) (0.4.6)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
csir_prosthetic\appdata\roaming\python\python313\site-packages (from
pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from pandas->datasets) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\
csir_prosthetic\appdata\local\programs\python\python313\lib\site-
packages (from pandas->datasets) (2025.1)
Requirement already satisfied: six>=1.5 in c:\users\csir_prosthetic\
appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas->datasets) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
training_args = TrainingArguments(
    output_dir="./results",
    evaluation_strategy="epoch",
    save_strategy="epoch",
    learning_rate=5e-5,
    per_device_train_batch_size=16, # Increased batch size
    per_device_eval_batch_size=16,
    num_train_epochs=50,
    weight_decay=0.01,
    save_total_limit=2,
    logging_dir="./logs",
    logging_steps=1000, # Reduce logging frequency
    fp16=True, # Enable mixed precision
    gradient_accumulation_steps=4, # Simulate larger batch
    report_to="none"
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_encodings,
    eval_dataset=test_encodings,
    tokenizer=tokenizer,
    data_collator=data_collator
)
```

```
C:\Users\CSIR_prosthetic\AppData\Local\Temp\
ipykernel_26068\4115539513.py:19: FutureWarning: `tokenizer` is
deprecated and will be removed in version 5.0.0 for
```

```
`Trainer.__init__`. Use `processing_class` instead.  
trainer = Trainer(  

```

```
trainer.train()  

```

```
<IPython.core.display.HTML object>  

```

```
TrainOutput(global_step=28650, training_loss=2.11125999570517,  
metrics={'train_runtime': 98233.6938, 'train_samples_per_second':  
18.689, 'train_steps_per_second': 0.292, 'total_flos':  
4.79514775781376e+17, 'train_loss': 2.11125999570517, 'epoch':  
49.99869281045751})  

```

```
import matplotlib.pyplot as plt  

```

```
losses = trainer.state.log_history  

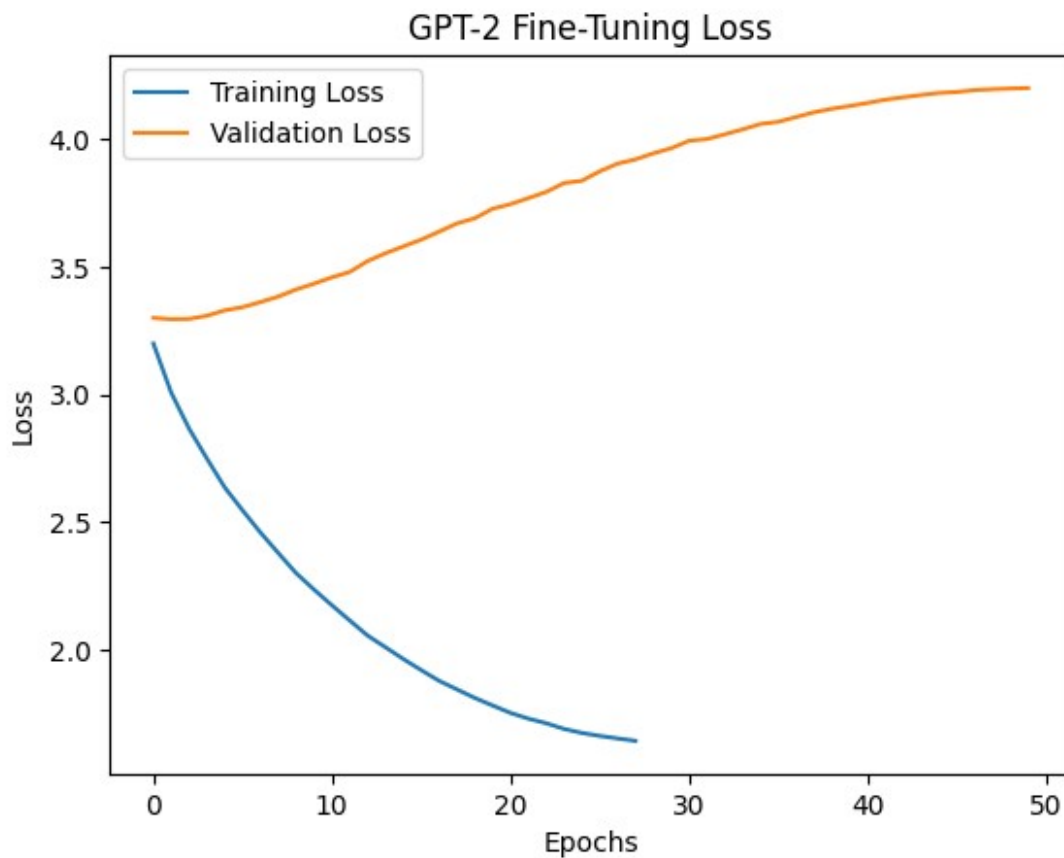
```

```
train_loss = [x["loss"] for x in losses if "loss" in x]  
eval_loss = [x["eval_loss"] for x in losses if "eval_loss" in x]  

```

```
plt.plot(train_loss, label="Training Loss")  
plt.plot(eval_loss, label="Validation Loss")  
plt.xlabel("Epochs")  
plt.ylabel("Loss")  
plt.title("GPT-2 Fine-Tuning Loss")  
plt.legend()  
plt.show()  

```



```
import math
```

```
perplexity = math.exp(eval_loss[-1])  
print(f"Model Perplexity: {perplexity:.2f}")
```

Model Perplexity: 66.64

```
model.eval()  
input_text = "The future of artificial intelligence is"  
input_ids = tokenizer.encode(input_text,  
    return_tensors="pt").to(device)  
output = model.generate(input_ids, max_length=50, temperature=1.0,  
    do_sample=True)  
print("Before Fine-Tuning:", tokenizer.decode(output[0],  
    skip_special_tokens=True))
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may

observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

Before Fine-Tuning: The future of artificial intelligence is still beset by the question of how and when humans will use artificial intelligence . AI organizations , by definition , are self @-@ sufficient . They simply do not know when the human level of intelligence has reached its limits

```
fine_tuned_model = GPT2LMHeadModel.from_pretrained("gpt2").to(device)
fine_tuned_model.eval()
```

```
GPT2LMHeadModel(
  (transformer): GPT2Model(
    (wte): Embedding(50257, 768)
    (wpe): Embedding(1024, 768)
    (drop): Dropout(p=0.1, inplace=False)
    (h): ModuleList(
      (0-11): 12 x GPT2Block(
        (ln_1): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (attn): GPT2Attention(
          (c_attn): Conv1D(nf=2304, nx=768)
          (c_proj): Conv1D(nf=768, nx=768)
          (attn_dropout): Dropout(p=0.1, inplace=False)
          (resid_dropout): Dropout(p=0.1, inplace=False)
        )
        (ln_2): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (mlp): GPT2MLP(
          (c_fc): Conv1D(nf=3072, nx=768)
          (c_proj): Conv1D(nf=768, nx=3072)
          (act): NewGELUActivation()
          (dropout): Dropout(p=0.1, inplace=False)
        )
      )
    )
    (ln_f): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
  )
  (lm_head): Linear(in_features=768, out_features=50257, bias=False)
)
```

```
fine_tuned_model.save_pretrained('./results')
tokenizer.save_pretrained('./results')
```

```
('./results\\tokenizer_config.json',
 './results\\special_tokens_map.json',
 './results\\vocab.json',
 './results\\merges.txt',
 './results\\added_tokens.json')
```

```
fine_tuned_model =
GPT2LMHeadModel.from_pretrained('./results').to(device)
```

```

tokenizer = GPT2Tokenizer.from_pretrained('./results')
fine_tuned_model.eval()

GPT2LMHeadModel(
  (transformer): GPT2Model(
    (wte): Embedding(50257, 768)
    (wpe): Embedding(1024, 768)
    (drop): Dropout(p=0.1, inplace=False)
    (h): ModuleList(
      (0-11): 12 x GPT2Block(
        (ln_1): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (attn): GPT2Attention(
          (c_attn): Conv1D(nf=2304, nx=768)
          (c_proj): Conv1D(nf=768, nx=768)
          (attn_dropout): Dropout(p=0.1, inplace=False)
          (resid_dropout): Dropout(p=0.1, inplace=False)
        )
        (ln_2): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
        (mlp): GPT2MLP(
          (c_fc): Conv1D(nf=3072, nx=768)
          (c_proj): Conv1D(nf=768, nx=3072)
          (act): NewGELUActivation()
          (dropout): Dropout(p=0.1, inplace=False)
        )
      )
    )
    (ln_f): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
  )
  (lm_head): Linear(in_features=768, out_features=50257, bias=False)
)

# Define input text
input_text = "In today's news, a significant event occurred where"

input_ids = tokenizer.encode(input_text,
return_tensors="pt").to(device)

# Create attention mask
attention_mask = (input_ids != tokenizer.pad_token_id).long()

# Generate output
output = fine_tuned_model.generate(input_ids,
                                   max_length=50,
                                   temperature=1.0,
                                   do_sample=True,
                                   top_p=0.9, # Nucleus sampling
                                   attention_mask=attention_mask,

pad_token_id=tokenizer.eos_token_id)

```

```
# Decode and print the generated text
```

```
print("Generated text: ", tokenizer.decode(output[0],  
skip_special_tokens=True))
```

Generated text: In today's news, a significant event occurred where people began to have an opinion about the importance of an effective policy in a country that is not in war.

On Sept. 7, President Obama called a meeting of Congress to discuss ways to

```
import torch  
from math import exp
```

```
# Calculate perplexity
```

```
output = fine_tuned_model(input_ids, labels=input_ids)  
loss = output.loss  
perplexity = exp(loss.item())  
print("Perplexity: ", perplexity)
```

Perplexity: 3505.6536162567

```
# Define input text
```

```
input_text = "In today's news, a significant event occurred where"
```

```
input_ids = tokenizer.encode(input_text,  
return_tensors="pt").to(device)
```

```
# Create attention mask
```

```
attention_mask = (input_ids != tokenizer.pad_token_id).long()  
output = fine_tuned_model.generate(input_ids,  
                                   max_length=100,  
                                   temperature=0.7,  
                                   do_sample=True,  
                                   top_p=0.9,  
                                   attention_mask=attention_mask,
```

```
pad_token_id=tokenizer.eos_token_id)
```

```
# Decode and print the generated text
```

```
print("Generated text: ", tokenizer.decode(output[0],  
skip_special_tokens=True))
```

Generated text: In today's news, a significant event occurred where the United States had been involved in a chemical attack on the Syrian town of Khan Shaykhun. The US government has said that the chemical attack was launched from Syria, not Iraq, which is believed to have carried out the attack.

The US-led coalition's "black helicopters" were ordered to assist the Syrian government in the fight against the Islamic State in Iraq and the Levant (ISIL), the United States and its allies said.

```
import torch
from transformers import GPT2LMHeadModel, GPT2Tokenizer

# Load the fine-tuned model and tokenizer
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
fine_tuned_model =
GPT2LMHeadModel.from_pretrained('./results').to(device)
tokenizer = GPT2Tokenizer.from_pretrained('./results')

# Ensure the model is in evaluation mode
fine_tuned_model.eval()

# Define input text (you can modify this)
input_text = "In a distant future, a new peace treaty was signed
between the nations of Mars and Earth. The treaty aimed to"

# Tokenize input text
input_ids = tokenizer.encode(input_text,
return_tensors="pt").to(device)

# Create attention mask (to handle padding correctly)
attention_mask = (input_ids != tokenizer.pad_token_id).long()

# Generate output using fine-tuned model
output = fine_tuned_model.generate(input_ids,
max_length=100, # You can adjust
the length
temperature=1.0, # Increase
randomness for more diverse output
do_sample=True, # Use sampling
top_p=0.9, # Nucleus
sampling (top-p)
top_k=50, # Limit to top 50
most likely tokens
attention_mask=attention_mask, #
Include attention mask

pad_token_id=tokenizer.eos_token_id) # Set pad token ID to EOS token

# Decode and print the generated text
generated_text = tokenizer.decode(output[0], skip_special_tokens=True)
print("Generated text: ", generated_text)
```

Generated text: In a distant future, a new peace treaty was signed between the nations of Mars and Earth. The treaty aimed to secure the protection of the planet from all known forms of warfare by

establishing a peaceful planet-wide peace treaty, but only the nations could agree to a peace treaty for the next 20 years. With a world divided between the powers, there was little opportunity for peaceful coexistence.

However, with the cooperation of a handful of powerful nations and two major empires, the two major powers

```
import torch
from transformers import GPT2LMHeadModel, GPT2Tokenizer

# Load the fine-tuned model and tokenizer
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
fine_tuned_model =
GPT2LMHeadModel.from_pretrained('./results').to(device)
tokenizer = GPT2Tokenizer.from_pretrained('./results')

# Ensure the model is in evaluation mode
fine_tuned_model.eval()

# Define input text (you can modify this)
input_text = input_text = "In a distant future, after the Mars-Earth
peace treaty was signed, a young diplomat named Sarah discovered a
secret plot that could shatter the peace. The plot involved"

# Tokenize input text
input_ids = tokenizer.encode(input_text,
return_tensors="pt").to(device)

# Create attention mask (to handle padding correctly)
attention_mask = (input_ids != tokenizer.pad_token_id).long()

# Generate output using fine-tuned model
output = fine_tuned_model.generate(input_ids,
max_length=100, # You can adjust
the length
temperature=1.0, # Increase
randomness for more diverse output
do_sample=True, # Use sampling
top_p=0.9, # Nucleus
sampling (top-p)
top_k=50, # Limit to top 50
most likely tokens
attention_mask=attention_mask, #
Include attention mask
pad_token_id=tokenizer.eos_token_id) # Set pad token ID to EOS token
# Decode and print the generated text
```

```
generated_text = tokenizer.decode(output[0], skip_special_tokens=True)
print("Generated text: ", generated_text)
```

Generated text: In a distant future, after the Mars-Earth peace treaty was signed, a young diplomat named Sarah discovered a secret plot that could shatter the peace. The plot involved the use of the U.S. military to infiltrate an ancient civilization's planet-building complex, the Mabuk, through a series of experiments, and take control of a human-run space station known as the S-4-S Enterprise. As the U.S. military began to discover more about her, Sarah decided